

Operating System Concepts

- OS Concepts
- Linux commands
- Shell scripts

Learning OS

- step 1: End user
 - Linux commands
- step 2: Administrator
 - Install OS (Linux)
 - Configuration - Users, Networking, Storage, ...
 - Shell scripts
- step 3: Programmer
 - Linux System call programming
- step 4: Designer/Internals
 - UNIX & Linux internals

What is OS?

- Interface between end user and computer hardware.
- Interface between Programs and computer hardware.
- Control program that controls execution of all other programs.
- Resource manager/allocator that manage all hardware resources.
- Bootable CD/DVD = Core OS + Applications + Utilities
- Core OS = Kernel -- Performs all basic functions of OS.

OS Functions

- CPU scheduling
- Process Management
- Memory Management
- File & IO Management
- Hardware abstraction
- User interfacing
- Security & Protection
- Networking

Disk/partition naming conventions

- * Windows:
 - * Disks are named as disk0, disk1, ...
 - * partitions are named as drives i.e. C:, D:, E:, ...
- * Linux:
 - * Disks are named as /dev/sda, /dev/sdb, /dev/sdc, etc.

- * Partitions per disk are named as
 - * sda partitions: sda1, sda2, sda3, ...
 - * sdb partitions: sdb1, ...

Path

- It is a unique location of any file in the file system.
- It is represented by character strings with few delimiters ("/", "\", ":")
- Types of path
 - There are two types of paths in linux
 - Absolute path
 - Path which starts with "/" is called as absolute path.
 - E.g. /home/devendra/MyData/Demos/demo01.sh
 - Relative path
 - Path with respect to current directory is called as relative path
 - E.g. MyData/Assignments/assign02.pdf

User interfacing

- UI of OS is a program (Shell) that interface between End user and Kernel.
- Shell -- Command interpreter
 - End user --> Command --> Shell --> Kernel
- User interfacing (Shell)
 - Graphical User Interface (GUI)
 - Command Line Interface (CLI)

Example shells

- Windows
 - GUI shell: explorer.exe
 - CLI shell: cmd.exe, powershell.exe
- DOS
 - CLI shell: command.com
- Unix/Linux
 - CLI shell: bsh, "bash", ksh, csh, zsh, ...
 - ls /bin/*sh
 - echo \$SHELL
 - shell of current user can be changed using "chsh" command.
- GUI shell/standards
 - GNOME: GNU Network Object Model Environment (e.g. Ubuntu, Redhat, CentOS, ...)
 - KDE: Kommon Desktop Environment (e.g. Kubuntu, SuSE, ...)